

Lecture 19 Notes: Stock Valuation and the Dividend Discount Model

Corporate Finance

I. Introduction: The Fascination with Stocks

Real-World Example: UBL Stocks

- An investment of **10 million rupees** in UBL stocks in 2021 would be worth approximately **135 million rupees** in 2026.
- Stock prices can appreciate significantly, making the stock market fascinating.
- However, values can also drop significantly.
- **Key Lesson:** It is vital to identify **good stocks and good businesses**, not just good companies.

II. Stocks vs. Bonds: Key Differences

Bonds (Fixed-Income)	Stocks (Equity)
Fixed maturity date	No maturity (can be held indefinitely)
Contractual cash flows (fixed coupons)	Dividends are NOT contractual
Bondholders are creditors	Stockholders are owners
No voting rights	Voting rights (elect board of directors)
Priority claim in liquidation	Residual owners (paid last)
Fixed return regardless of company performance	Return depends on firm earnings

Key Characteristics of Stocks

- When you buy stocks, you buy **ownership** in the company's net assets (assets minus liabilities).
- Your **liability is limited** to the amount of your individual investment.
- Stockholders have voting rights to elect the board of directors and participate in decision-making.
- If the company incurs losses, you may not receive any dividend.

III. Control, Proxies, and Takeovers

Proxy Voting

- Shareholders who cannot attend meetings can use a **proxy form** to appoint a representative to vote on their behalf.
- Management or influential stockholders gather proxies to attain control or change management.
- **Takeover Example in Pakistan:** Adamjee Insurance was taken over through a proxy fight.

Preemptive Rights

- Common stockholders have **preemptive rights** to purchase new shares on a pro-rata basis.
- This allows them to maintain their ownership percentage when new shares are issued.

IV. Corporate Goals and Agency Problems

Primary Goal

The primary goal of management should be the **maximization of shareholder wealth** by maximizing the stock price.

Agency Problems

- Managers are agents for shareholders.
- **Agency problems** occur when managers engage in activities that do not align with shareholder objectives.
- In the US, "Say on Pay" laws allow shareholders to vote on CEO compensation to ensure resources are used correctly.

V. Classes of Stock

Some companies have multiple classes of stock with different voting rights.

Example: Google (Alphabet)

Share Class	Voting Rights
Class A Shares	1 vote per share
Class B Shares	10 votes per share

Class B shares are held by the two founders and the CEO to **protect their ownership and control**.

VI. Price vs. Intrinsic Value

Definitions

- **Price (P_0):** What the market is currently paying for a stock (directly observable).
- **Intrinsic Value ($\hat{P}_0$):** The **true value** of the stock justified by company fundamentals (must be estimated).

Market Equilibrium

Market Equilibrium occurs when $P_0 = \hat{P}_0$

Investment Decision Rules

Condition	Action
$P_0 < \hat{P}_0$ (Undervalued)	BUY
$P_0 > \hat{P}_0$ (Overvalued)	SELL
$P_0 = \hat{P}_0$ (Fairly Valued)	HOLD

Why Prices and Intrinsic Values Differ

- **Perceived risk** may differ from **true risk**.
- Managerial actions, economic environment, taxes, and political climate affect perceptions.

Example: 2008 Subprime Mortgage Crisis

- Before the crisis, bank stocks were highly rated (low perceived risk).
- **True risk was much higher** due to huge investments in mortgage-backed securities.
- When property values dropped, perceptions changed and stock prices plummeted.
- **Bear Stearns:** Fell from \$171 per share (2007) to just \$2 per share (March 2008).

VII. The Dividend Discount Model (DDM)

Basic Principle of Asset Pricing

The price of any asset today is the **present value of all future expected cash flows**.

$$\hat{P}_0 = \sum_{t=1}^{\infty} \frac{E(\text{Cash Flow}_t)}{(1+r)^t}$$

Cash Flows for a Stock

For a stock, the expected cash flows are:

1. **Dividends** (periodic cash distributions)
2. **Capital Gain** (the price at which you eventually sell the stock)

If You Hold the Stock Forever

$$\hat{P}_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r_s)^t}$$

Where:

- D_t = Expected dividend at time t
- r_s = Required return on the stock

VIII. Constant Growth (Gordon Growth) Model

Assumptions

1. Dividends grow at a **constant rate** (g) forever.
2. The growth rate is **less than the required return** ($g < r_s$).
3. The required return (r_s) is calculated using the CAPM:

$$r_s = r_{RF} + \beta \times (r_m - r_{RF})$$

The Gordon Growth Model Formula

$$\hat{P}_0 = \frac{D_1}{r_s - g}$$

Where:

- $\hat{P}_0$ = Intrinsic value of the stock today
- D_1 = Expected dividend next year = $D_0 \times (1 + g)$

- r_s = Required return
- g = Constant growth rate
- **Condition:** $g < r_s$ (for the model to work mathematically and logically)

Example Calculation

Given:

$$D_0 = \$2.00 \quad (\text{Current dividend})$$

$$g = 6\% = 0.06$$

$$r_s = 13\% = 0.13$$

Step 1: Calculate the expected dividend next year (D_1)

$$D_1 = D_0 \times (1 + g) = 2.00 \times 1.06 = \$2.12$$

Step 2: Apply the Gordon Growth Model

$$\hat{P}_0 = \frac{D_1}{r_s - g} = \frac{2.12}{0.13 - 0.06} = \frac{2.12}{0.07} = \$30.29$$

Answer

$$\hat{P}_0 = \$30.29$$

IX. Interpreting the Gordon Growth Model

Key Insights

- The model shows that stock value is determined by three factors:
 1. **Current dividend** (D_0) - higher dividends increase value
 2. **Growth rate** (g) - higher growth increases value
 3. **Required return** (r_s) - higher required return decreases value
- The condition $g < r_s$ is essential. If $g \geq r_s$, the model breaks down (would give infinite or negative values).
- The model assumes dividends grow at a constant rate forever - a strong assumption but useful for mature, stable companies.

X. Variations of the DDM

In future classes, we will explore:

- **Non-constant growth** (multistage growth models)

- High growth for several years, then constant growth thereafter
- Example: A company growing at 15% for 5 years, then 5% forever
- **Zero growth** ($g = 0$): $\hat{P}_0 = D/r_s$ (perpetuity formula)
- **Corporate Valuation Model** (Free Cash Flow approach)
- **P/E multiples** approach

XI. Final Syllabus

Remaining Topics

- **Chapter 9:** Stock Valuation (current topic)
- **Chapter 10:** Cost of Capital
- Handouts on **Sources of Financing**
- Handouts on **Financial Markets**

Formula Summary Sheet

Key Formulas

$$\text{Gordon Growth Model : } \hat{P}_0 = \frac{D_1}{r_s - g} \quad (1)$$

$$\text{Expected Dividend : } D_1 = D_0 \times (1 + g) \quad (2)$$

$$\text{CAPM (Required Return) : } r_s = r_{RF} + \beta(r_m - r_{RF}) \quad (3)$$

$$\text{Zero Growth Model : } \hat{P}_0 = \frac{D}{r_s} \quad (4)$$

$$\text{Basic DDM : } \hat{P}_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + r_s)^t} \quad (5)$$